DIFFERENTIAL EQUATION

- **Differential Equation**: - An equation involving one dependent variable and its derivatives with respect to one or more independent variable is called as differential equation.

- **Ordinary differential equation**: - An ordinary differential equation is one in which there is only one independent variable, so that all the derivatives occurring in it are ordinary derivatives.

- **Partial differential equation**: - A Partial differential equation is one involving more than one independent variable, so that all the derivatives occurring in it are partial derivatives.

- **Order of differential equation**: - The order of the differential equation is the order of the highest derivatives present.

- **Degree of differential equation**: - The degree of differential equation is the exponent of highest order derivatives present in a differential equation after the differential equation has been rationalized and made free of fractional powers with regards to all derivatives present.
- **First order differential equation**: -An equation contains $\frac{dy}{dx}$, y and function of x is called as first order differential equation.

- **Solution of differential equation**: - Any function free from derivatives, satisfying the differential equation identically is said to be a solution of differential equation.

- **General Solution of Differential Equation**: The solution of differential equation which contains a number of arbitrary constants equals to the order of the differential equation is called as general solution. Hence the general solution of a differential equation of nth degree has n number of arbitrary constants.

- **Particular Solution**: A solution obtained by giving particular values to arbitrary constants in the general solution is called a particular solution.

- **Singular solution**: -A differential equation may sometimes have an additional solution that cannot be obtained from the general solution and is then called a singular solution. It does not contain any arbitrary constants nor can be derived from the complete solution giving any particular values to the arbitrary constants.

- **n^{th} order ordinary differential equation**: - An n^{th} order differential equation in general form is $F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, \ldots\ldots\ldots\ldots\ldots\ldots, \frac{d^ny}{dx^n}\right) = 0 ------------\rightarrow (1)$
  Solution of (1) containing arbitrary constant is called as general solution.
  Any solution obtained from the general solution by giving a particular value to the arbitrary constants is known as particular solution.

- **Initial Value problem**: - A differential equation together with an initial condition is known as initial value problem, which can be expressed in the form of: -
  $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$. It means at x = x_0, y = y_0

- **Method of solving first order and first degree differential equation:**

  - **Variable Separable Method:** If a differential equation can be expressed in the form of $\frac{dy}{dx} = \frac{f(x)}{g(y)}$ or $g(y)dy = f(x)dx$ then it is said to be of type variable separable and such equation can be solved by integrating both sides. So solution given by it is $\int g(y)dy = \int f(x)dx + C$, where C is any arbitrary constant.

  - **Equation Reducible to Separable Form:** The differential equation of the form $\frac{dy}{dx} = f(ax + by + c)$ can be reduced to variable separable from by substituting $ax + by + c = t$.
    $\Rightarrow a + b\frac{dy}{dx} = \frac{dt}{dx}$
    $\Rightarrow a + bf(t) = \frac{dt}{dx}$
    $\Rightarrow \frac{dt}{a + bf(t)} = dx$ reduced to separable form.

  - **Homogeneous Equation:** If the given differential equation of the form $\frac{dy}{dx} = f(x, y)$ and $f(tx, ty) = t^n f(x, y)$ then it is called as homogeneous differential equation. Or if it is in the form of $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$ or $\frac{dy}{dx} = f\left(\frac{x}{y}\right)$ it is called as homogeneous differential equation.
    To solve this put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ in the given differential equation and then solve by variable separable method.

  - **If the differential equation of the form** $\frac{dy}{dx} = f(ax + by + c)$. Put $ax + by + c = t$ in the given differential equation and solve.

  - **Non Homogeneous Equation:** If the given differential equation of the form $\frac{dy}{dx} = \frac{Ax + By + C}{ax + by + c}$ where A, B, C, a, b, c are constants, then it is called as non-homogeneous differential equation.

(a) If $\frac{A}{a}=\frac{B}{b}$, then put $ax+by+c=t$ and solve the differential equation.

(b) If $\frac{A}{a}\neq\frac{B}{b}$, then put $x=X+h$ and $y=Y+k$ we get $\frac{dY}{dX}=\frac{AX+BY+(Ah+Bk+C)}{aX+bY+(ah+bk+c)}$. Then put $Ah+Bk+C=0,\ ah+bk+c=0$ and solve to find the values of h,k. The equation now can be converted to homogeneous differential equation as $\frac{dY}{dX}=\frac{AX+BY}{aX+bY}$. Solve the same and find the solution as $Y=F(X)$. Then put $X=x-h,\ Y=y-k$ to get the required solution.

- **<u>Linear Differential Equation</u>**: - A first order differential equation is said to be linear if it is in the form of $y'+p(x)y=r(x)$.------------→(1)

  Its integrating factor $I.F.=e^{\int pdx}$.
  Multiplying I.F. both side of equation (1) we get
  $\Rightarrow y'e^{\int pdx}+p(x)e^{\int pdx}y=r(x)e^{\int pdx}$
  $\Rightarrow \frac{d}{dx}\left(ye^{\int pdx}\right)=r(x)e^{\int pdx}$
  On integrating both side we get
  $\Rightarrow \left(ye^{\int pdx}\right)=\int r(x)e^{\int pdx}dx$

- **<u>Linear Differential Equation</u>**: - A first order differential equation is said to be linear if it is in the form of $x'+p(y)x=r(y)$.------------→(1)

  Its integrating factor $I.F.=e^{\int pdy}$.
  Multiplying I.F. both side of equation (1) we get
  $\Rightarrow x'e^{\int pdy}+p(y)e^{\int pdy}x=r(y)e^{\int pdy}$
  $\Rightarrow \frac{d}{dy}\left(xe^{\int pdy}\right)=r(y)e^{\int pdy}$
  On integrating both side we get
  $\Rightarrow xe^{\int pdy}=\int r(y)e^{\int pdy}dy$

- **<u>Reduction to linear differential equation (Bernoulli's differential equation)</u>** : - A first order differential equation is said to be Bernoulli's differential equation if it is in the form of $y'+p(x)y=r(x)y^n$------------→(1)

  Divide the equation (1) by y^n then equation (1) becomes
  $\Rightarrow y^{-n}y'+p(x)y^{1-n}=r(x)$-------------→(2)

Put $y^{1-n}=t \Rightarrow (1-n)y^{-n}y'=t' \Rightarrow y^{-n}y'=\frac{t'}{(1-n)}$, then equation (2) becomes

$\Rightarrow \frac{t'}{1-n}+p(x)t=r(x)$

$\Rightarrow t'+(1-n)p(x)t=(1-n)r(x)$ which is first order linear differential equation.

- **Exact Differential equation**: A first order differential equation of the form $M(x,y)dx+N(x,y)dy=0$ is called as exact differential equation if $\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$. Its solution is given by $\int\limits_{y-\text{cons}\tan t} Mdx+\int\limits_{Not-containing-x-terms} Ndy=c$.
- **Integrating Factor**: If an equation $M(x,y)dx+N(x,y)dy=0$ is not exact. Then it can be made exact by multiplying some functions of x and y. Such a function is called as Integrating Factor.

- **Solution of Exact Differential equation:**

- **By Inspection Method**:

(1) $dx+dy=d(x+y)$

(2) $ydx+xdy=d(xy)$

(3) $\frac{ydx-xdy}{y^2}=d\left(\frac{x}{y}\right)$

(4) $\frac{xdy-ydx}{x^2}=d\left(\frac{y}{x}\right)$

(5) $\frac{2xydx-x^2dy}{y^2}=d\left(\frac{x^2}{y}\right)$

(6) $\frac{2xydy-y^2dx}{x^2}=d\left(\frac{y^2}{x}\right)$

(7) $\frac{2xy^2dx-2x^2ydy}{y^4}=d\left(\frac{x^2}{y^2}\right)$

(8) $\frac{2x^2ydy-2xy^2dx}{x^4}=d\left(\frac{y^2}{x^2}\right)$

(9) $2(xdx+ydy)=d(x^2+y^2)$

(10) $\frac{ydx+xdy}{xy}=\frac{dx}{x}+\frac{dy}{y}=d(\log(xy))$

(11) $\frac{ydx-xdy}{xy}=\frac{dx}{x}-\frac{dy}{y}=d\left(\log\left(\frac{x}{y}\right)\right)$

(12) $\dfrac{xdy - ydx}{xy} = \dfrac{dy}{y} - \dfrac{dx}{x} = d\left(\log\left(\dfrac{y}{x}\right)\right)$

(13) $\dfrac{xdy - ydx}{x^2 + y^2} = d\left(\tan^{-1}\left(\dfrac{y}{x}\right)\right)$

(14) $\dfrac{ydx - xdy}{x^2 + y^2} = d\left(\tan^{-1}\left(\dfrac{x}{y}\right)\right)$

(15) $\dfrac{xdx + ydy}{x^2 + y^2} = d\left(\dfrac{1}{2}\log\left(x^2 + y^2\right)\right)$

(16) $\dfrac{xdx + ydy}{\sqrt{x^2 + y^2}} = d\left(\sqrt{x^2 + y^2}\right)$

(17) $\dfrac{xdy + ydx}{x^2 y^2} = d\left(\dfrac{-1}{xy}\right)$

(18) $\dfrac{ye^x dx - e^x dy}{y^2} = d\left(\dfrac{e^x}{y}\right).$

(19) $\dfrac{xe^y dy - e^y dx}{x^2} = d\left(\dfrac{e^y}{x}\right)$

(20) $x^{m-1} y^{n-1}(mydx + nxdy) = d\left(x^m y^n\right)$

- If the given equation $M(x,y)dx + N(x,y)dy = 0$ is homogeneous and $Mx + Ny \neq 0$, then $\dfrac{1}{Mx + Ny}$ is an integrating factor.
- If the given equation $M(x,y)dx + N(x,y)dy = 0$ is homogeneous and $Mx - Ny \neq 0$, then $\dfrac{1}{Mx - Ny}$ is an integrating factor.
- If $\dfrac{\dfrac{\partial M}{\partial y} - \dfrac{\partial N}{\partial x}}{N} = f(x)$ then integrating factor is $e^{\int f(x)dx}$.
- If $\dfrac{\dfrac{\partial N}{\partial x} - \dfrac{\partial M}{\partial y}}{M} = f(y)$ then integrating factor is $e^{\int f(y)dy}$.

- If the equation $M(x,y)dx + N(x,y)dy = 0$ is of the form $x^{\tilde{}}\, y \;\; (mydx + nxdy) = 0$, then its integrating factor is $x^{km-1-\sim}\, y^{nk-1-}$, where k can have any value.

- **<u>Orthogonal Trajectory</u>**: Any curve which cuts every member of a given family of curves at right angle is called an orthogonal trajectory of the family.

- **Process to find the Orthogonal trajectory**:

(a) Let $f(x, y, c) = 0$ be the equation, where c is any arbitrary parameter.
(b) Differentiate the given equation with respect to x and then eliminate c.
(c) Replace

- **Solution of Differential Equation of first Degree but not of First Degree**: Differential equation of first order and nth degree is $p^n + f_1 p^{n-1} + f_2 p^{n-2} + f_3 p^{n-3} + ... + f_{n-1} p^1 + f_n = 0$, where $p = \frac{dy}{dx}$ and $f_1, f_2, f_3, ..., f_{n-1}, f_n$ are functions of x and y. Its general solution contains only one arbitrary constants as it is of first order differential equation.
- **Different Types of Differential Equation of first Degree but not of First Degree**:

(a) **Equation Solvable for p**:
Let us solve $p^n + f_1 p^{n-1} + f_2 p^{n-2} + f_3 p^{n-3} + ... + f_{n-1} p^1 + f_n = 0$
Resolving the LHS as
$\Rightarrow (p - g_1)(p - g_2)(p - g_3)...(p - g_n) = 0$
$\Rightarrow (p - g_1) = 0, (p - g_2) = 0, (p - g_3). = 0, \quad ..(p - g_n) = 0$
$\Rightarrow t_1(x, y, c) = 0, t_2(x, y, c) = 0, t_3(x, y, c). = 0, \quad ..t_n(x, y, c) = 0$
$\Rightarrow t_1(x, y, c) \times t_2(x, y, c) \times t_3(x, y, c) \times ... \times t_n(x, y, c) = 0$ is the solution.

(b) **Equation Solvable for y:**
Let the equation can be written in the form $y = f(x, p)$, then it is solvable for y.
Differentiate this equation with respect to x, then we get
$\Rightarrow ¤(x, p, c) = 0$
The eliminate p between this and the original equation, which gives relation between x, y and c, which is the required solution.
(c) **Equation Solvable for x:**
Let the equation can be written in the form $x = f(y, p)$, then it is solvable for x.
Differentiate this equation with respect to y, then we get
$\Rightarrow ¤(y, p, c) = 0$
The eliminate p between this and the original equation, which gives relation between x, y and c, which is the required solution.
(d) **Clairut's Equation**: Differential Equation of the form $y = px + f(p)$ is known as Clairut's equation.
$y = px + f(p)$ --------------→(1)
Differentiating (1) with respect to x, we get
$\Rightarrow p = p + x\frac{dp}{dx} + f'(p)\frac{dp}{dx}$
$\Rightarrow (x + f'(p))\frac{dp}{dx} = 0$

$$\Rightarrow (x + f'(p)) = 0, \qquad \frac{dp}{dx} = 0$$

If $\frac{dp}{dx} = 0$, then p is constant c.

Eliminating p from (1) by putting $p = c$ we get

$y = cx + f(c)$ is the general solution.

And if $(x + f'(p)) = 0$, then by eliminating p we will get another solution which is called as singular solution.

- Sometimes transformation to the polar co-ordinates facilitates separation of variables. In this connection it is convenient to remember the following:

(a) If $x = r\cos\theta$, $y = r\sin\theta$, then

(i) $xdx + ydy = rdr$.

(ii) $xdx - ydy = r^2 d\theta$.

(iii) $dx^2 + dy^2 = dr^2 + r^2 d\theta^2$.

(b) If $x = r\sec\theta$, $y = r\tan\theta$, then

(i) $xdx - ydy = rdr$.

(ii) $xdy - ydx = r^2 \sec\theta d\theta$

- **<u>Solution of differential equation of higher degree</u>**:

- **<u>Linear Differential Equation</u>**: - A differential equation in which the dependent variable & its derivatives occur only in the first degree & are not multiplied together is known as Linear Differential equation.

  General form of Linear Differential Equation of nth order is:

  $\frac{d^n y}{dx^n} + P_1 \frac{d^{n-1} y}{dx^{n-1}} + P_2 \frac{d^{n-2} y}{dx^{n-2}} + ... + P_n y = X$, where $P_1, P_2, ..., P_n, X$ are functions of x.

- **<u>Linear Differential Equation With Constant Coefficients</u>**: - In general form of Linear Differential Equation of nth order if $P_1, P_2, ..., P_n$ all are replaced by constants $a_1, a_2, ..., a_n$ then equation becomes $\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + ... + a_n y = X$ is called as Linear Differential Equation with constant coefficients.

- **<u>Homogeneous / Non Homogeneous Linear Differential Equation With Constant Coefficients</u>**: -

  $\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + ... + a_n y = X$ ----------------------------→(1)

  If $X = 0$, then equation (1) is known as Homogeneous Linear Differential Equation with constant coefficients.

If $X \neq 0$, then equation (1) is known as Non Homogeneous Linear Differential Equation with constant coefficients.

- **Solution of Linear Differential equation with Constant Coefficients: -**

Linear differential equation with constant coefficients has two solutions.

1. Complementary function ---------------→ Complete solution. ----→ C.F.
2. Particular Integral -----------→ Particular Solution ---→ P.I.
3. Solution = C.F. + P.I.

- **Operator D**: -

$D = \frac{d}{dx}$, $D^2 = \frac{d^2}{dx^2}$, $D^3 = \frac{d^3}{dx^3}$ ……………….., $D^n = \frac{d^n}{dx^n}$

Now Linear differential equation with constant coefficients

$\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + ... + a_n y = X$ can be written as by using operator

$\Rightarrow D^n y + a_1 D^{n-1} y + a_2 D^{n-2} y + ... + a_n y = X$

$\Rightarrow \left(D^n + a_1 D^{n-1} + a_2 D^{n-2} + ... + a_n\right) y = X$

which is known as symbolic form of Linear differential equation with constant coefficients.

- **Rules for finding Complementary Function (C.F.)**: -

Let us find complementary function of $\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + ... + a_n y = 0$ ----→(1)

Steps: -

1. Write given equation (1) in symbolic form.
   $\Rightarrow D^n y + a_1 D^{n-1} y + a_2 D^{n-2} y + ... + a_n y = 0$
   $\Rightarrow \left(D^n + a_1 D^{n-1} + a_2 D^{n-2} + ... + a_n\right) y = 0$ ----------------→(2)
2. Write its Auxiliary equation.
   $\Rightarrow D^n + a_1 D^{n-1} + a_2 D^{n-2} + ... + a_n = 0$ --------------→(3)
3. Find roots of equation (3). Let roots are $m_1, m_2, ..., m_n$.
4. If all roots are real and distinct, then solution:
   $C.F. = C_1 e^{m_1 x} + C_2 e^{m_2 x} + ... + C_n e^{m_n x}$
5. If two roots are equal & rest are distinct and all roots are real, then solution:
   $C.F. = (C_1 + C_2 x) e^{m_1 x} + C_3 e^{m_3 x} + ... + C_n e^{m_n x}$
6. If three roots are equal & rest are distinct and all roots are real, then solution:
   $C.F. = \left(C_1 + C_2 x + C_3 x^2\right) e^{m_1 x} + C_4 e^{m_4 x} + ... + C_n e^{m_n x}$
7. If one pair of imaginary roots $\sim \pm i$ and rest are real and distinct, then solution:
   $C.F. = (C_1 \cos \ \ x + C_2 \sin \ \ x) e^{\sim x} + C_3 e^{m_3 x} + ... + C_n e^{m_n x}$
8. If two pair of equal imaginary roots $\sim \pm i$, $\sim \pm i$ and rest are real and distinct, then solution:
   $C.F. = ((C_1 + C_2 x)\cos \ \ x + (C_3 + C_4 x)\sin \ \ x) e^{\sim x} + C_5 e^{m_5 x} + ... + C_n e^{m_n x}$

- **Inverse Operator**: - If $f(D)\left\{\frac{1}{f(D)}X\right\} = X$, then $f(D), \frac{1}{f(D)}$ are inverse operator to each other.
- **Theorem 1**: - $\frac{1}{D}X = \int X dx$

  Proof: - Let $\frac{1}{D}X = y$

  $$\Rightarrow D\left(\frac{1}{D}X\right) = D(y) \Rightarrow X = D(y) \Rightarrow y = \int X dx \Rightarrow \frac{1}{D}X = \int X dx \qquad \text{(Proved).}$$
- **Theorem 2**: - $\frac{1}{D-a}X = e^{ax}\int Xe^{-ax}dx$

  Proof: - Let $\frac{1}{D-a}X = y$

  $$\Rightarrow (D-a)\left(\frac{1}{D-a}X\right) = (D-a)(y) \Rightarrow X = Dy - ay$$

  $$\Rightarrow \frac{dy}{dx} - ay = X \text{, Here } I.F. = e^{-\int a dx} = e^{-ax}$$

  So, solution $y.e^{-ax} = \int Xe^{-ax}dx \Rightarrow y = e^{ax}\int Xe^{-ax}dx$

  $$\Rightarrow \frac{1}{D-a}X = e^{ax}\int Xe^{-ax}dx$$
- **Rules for finding Particular Integral**: -

  Let us find particular integral of $\frac{d^n y}{dx^n} + a_1\frac{d^{n-1}y}{dx^{n-1}} + a_2\frac{d^{n-2}y}{dx^{n-2}} + \dots\dots\dots + a_n y = X$

  Steps: -

  1. Write its symbolic form.

     $(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots\dots\dots + a_n)y = X$

     $\Rightarrow F(D)y = X, \qquad f(D) = (D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots\dots\dots + a_n)$
  2. $P.I. = \frac{1}{f(D)}X = \frac{1}{(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots\dots\dots + a_n)}X$

  Case 1: - If $X = e^{ax}$

  Then, $P.I. = \frac{1}{f(D)}e^{ax}$

  $\Rightarrow P.I. = \frac{e^{ax}}{f(a)}$ if $f(a) \neq 0$

  $\Rightarrow P.I. = \frac{xe^{ax}}{f'(a)}$ if $f(a) = 0$ and $f'(a) \neq 0$

$\Rightarrow P.I. = \dfrac{x^2 e^{ax}}{f''(a)}$ if $f(a) = 0$, $f'(a) = 0$ and $f''(a) \neq 0$

$\Rightarrow P.I. = \dfrac{x^3 e^{ax}}{f'''(a)}$ if $f(a) = 0$, $f'(a) = 0$, $f''(a) = 0$ and $f'''(a) \neq 0$

Similarly continue.

Case 2: - If $X = \sin(ax+b)$ or $X = \cos(ax+b)$

i. $P.I. = \dfrac{1}{f(D^2)}\sin(ax+b)$ or $P.I. = \dfrac{1}{f(D^2)}\cos(ax+b)$

$\Rightarrow P.I. = \dfrac{1}{f(-a^2)}\sin(ax+b)$ or $\Rightarrow P.I. = \dfrac{1}{f(-a^2)}\cos(ax+b)$, if $f(-a^2) \neq 0$.

ii. If $f(-a^2) = 0$

Then $P.I. = \dfrac{1}{f(D)}\sin(ax+b)$ or $P.I. = \dfrac{1}{f(D)}\cos(ax+b)$

$\Rightarrow P.I. = impart\left(\dfrac{1}{f(D)}e^{i(ax+b)}\right)$ or $\Rightarrow P.I. = repart\left(\dfrac{1}{f(D)}e^{i(ax+b)}\right)$

Case 3: - If $X = x^m$

Then, $P.I. = \dfrac{1}{f(D)}x^m$

$\Rightarrow P.I. = [f(D)]^{-1}x^m$,

Then represent $[f(D)]^{-1}$ as ascending power of D & then operate at x^m.

Case 4: - If $X = e^{ax}.V$

Then, $P.I. = \dfrac{1}{f(D)}e^{ax}.V$

$\Rightarrow P.I. = e^{ax}\dfrac{1}{f(D+a)}.V$

Case 5: - If X is any other function of x.

Then, $P.I. = \dfrac{1}{f(D)}X$

$$\Rightarrow P.I. = \frac{1}{(D-m_1)(D-m_2)\,.........\,(D-m_n)}X$$

$$\Rightarrow P.I. = \left(\frac{A_1}{D-m_1} + \frac{A_2}{D-m_2} + ... + \frac{A_n}{D-m_n}\right)X$$

$$\Rightarrow P.I. = A_1 e^{m_1 x}\int X e^{-m_1 x}dx + A_2 e^{m_2 x}\int X e^{-m_2 x}dx + ... + A_1 e^{m_3 x}\int X e^{-m_3 x}dx$$

By Using $\dfrac{1}{D-a}X = e^{ax}\int X e^{-ax}dx$

- **Solution of Non homogeneous Linear Differential Equation With Constant Coefficients**:

Let us solve $\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + ... + a_n y = X$

Steps: -

I. Write its symbolic form.
II. Write its Auxiliary equation.
III. Find Complementary Function C.F.
IV. Find Particular Integral P.I.
V. Solution $y = C.F. + P.I.$

- **Method of Variation of Parameter to find Particular Integral: -**

Let us solve $\frac{d^2 y}{dx^2} + p\frac{dy}{dx} + qy = X$ -------------→(1)

Steps: -

1. Write in symbolic form.
   $\left(D^2 + pD + q\right)y = X$ ---------------------→(2)
2. Write its Auxiliary equation.
   $D^2 + pD + q = 0$ ----------------→(3)
3. Solve equation (3) and find roots of (3). Let roots are $D = y_1, \; y_2$
   Then $C.F. = D = C_1 y_1 + C_2 y_2$ ------------------→(4)
4. Write $y_1, \; y_2, \; X$
5. Find Wronskian of $\left(y_1, y_2\right) = W\left(y_1, y_2\right) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = W$ -------→(5)
6. Then find $P.I. = y_p = -y_1 \int \frac{y_2 . X dx}{W} + y_2 \int \frac{y_1 . X dx}{W}$.------------→(6)
7. Then Solution $y = C.F. + P.I. = y_h + y_p$ -------------→(7)

- **Method of U to find Particular Integral: -**

It can be find by taking trial solution containing unknown constants which are determined by substituting in the given equation.
Let us solve $f(D)y = X$ ------------→(1)

Steps: -

1. Write its Auxiliary equation.
   $f(D) = 0$. Solve it and find roots.
2. Write its complementary function C.F.
3.
   i. If $X = ae^{-x}$,
      Then take $P.I. = ke^{-x}$ and put in the given equation and find the value of k.

ii. If $X = a_0 + a_1x + a_2x^2 + ... + a_nx^n$,

Then take $P.I. = A_0 + A_1x + A_2x^2 + ... + A_nx^n$, put in the given equation and find the value of $A_0, A_1, A_2, ..., A_n$.

iii. If $X = a\sin bx$ or $X = a\cos bx$,

Then take $P.I. = A\sin bx + B\cos bx$, put in the given equation and find the value of A and B.

iv. If $X = a\tan bx$ or $X = a\sec bx$

Then method fails.

4. Then solution $y = C.F. + P.I$

- **<u>Method of Undetermined Coefficient to find Particular Integral</u>: -**

It can be find by taking trial solution containing unknown constants which are determined by substituting in the given equation.

Let us solve $f(D)y = X$ -------------→(1)

<u>Steps</u>: -

1. Write its Auxiliary equation.
   $f(D) = 0$. Solve it and find roots.
2. Write its complementary function C.F.
3.

v. If $X = ae^{\sim x}$,

Then take $P.I. = ke^{\sim x}$ and put in the given equation and find the value of k.

vi. If $X = a_0 + a_1x + a_2x^2 + ... + a_nx^n$,

Then take $P.I. = A_0 + A_1x + A_2x^2 + ... + A_nx^n$, put in the given equation and find the value of $A_0, A_1, A_2, ..., A_n$.

vii. If $X = a\sin bx$ or $X = a\cos bx$,

Then take $P.I. = A\sin bx + B\cos bx$, put in the given equation and find the value of A and B.

viii. If $X = ae^{\sim x}\sin bx$ or $X = be^{\sim x}\cos bx$,

Then take $P.I. = e^{\sim x}\left(A\sin bx + B\cos bx\right)$, put in the given equation and find the value of A and B.

ix. If $X = a\tan bx$ or $X = a\sec bx$

Then method fails.

4. Then solution $y = C.F. + P.I$

- **<u>Cauchy's Linear Differential Equation</u>: -** A Differential equation of the form

$x^n\dfrac{d^ny}{dx^n} + a_1x^{n-1}\dfrac{d^{n-1}y}{dx^{n-1}} + a_2x^{n-2}\dfrac{d^{n-2}y}{dx^{n-2}} + ... + a_ny = X$ is known as Cauchy's Linear differential equation.

- **Solution of Cauchy's Linear Differential Equation: -**

Steps: -

1. Put $x = e^t \Rightarrow t = \log x$
2. Let $D = \frac{d}{dt}$ then put

   $$x\frac{dy}{dx} = Dy,\quad x^2\frac{d^2y}{dx^2} = D(D-1)y,\quad x^3\frac{d^3y}{dx^3} = D(D-1)(D-2)y, \ldots\ldots..$$

   in the given equation.
3. Then Cauchy Linear Differential Equation reduces to Linear Differential Equation with constant coefficients and hence solve as per the steps.

- **Legendre's Linear Differential Equation: -** A Differential equation of the form

  $(ax+b)^n\frac{d^ny}{dx^n} + a_1(ax+b)^{n-1}\frac{d^{n-1}y}{dx^{n-1}} + a_2(ax+b)^{n-2}\frac{d^{n-2}y}{dx^{n-2}} + \ldots\ldots.. + a_n y = X$ is known as Legendre's Linear differential equation.

- **Solution of Legendre's Linear Differential Equation: -**

  Steps: -

1. Put $ax+b = e^t \Rightarrow t = \log(ax+b)$
2. Let $D = \frac{d}{dt}$, then put

$$(ax+b)\frac{dy}{dx} = aDy,\quad (ax+b)^2\frac{d^2y}{dx^2} = a^2D(D-1)y,\quad (ax+b)^3\frac{d^3y}{dx^3} = a^3D(D-1)(D-2)y, \ldots\ldots..$$

in the given equation.

Then Legendre's Linear Differential Equation reduces to Linear Differential Equation with constant coefficients and hence solve as per the steps.

- **Simultaneous system of Linear Differential Equation with Constant Coefficients: -**

Let us solve $f_1(D)x + f_2(D)y = T_1(t)$ -------------→(1)

$g_1(D)x + g_2(D)y = T_2(t)$ -------------→(2)

Steps: -

1. This simultaneous system of linear differential equation can be solved as same as the system of linear equation.
2. Eliminate y.

   $(1)\times g_2(D) - (2)\times f_1(D)$ and then solve for x.
3. Put value of x in (1) or (2) and solve for y.

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